

Digital transformation: the role of generative AI in the evolution of knowledge management systems

Aline Luiza Brusco Pletsch, Graciele Tonial, Florinda Matos and Leani Lauermann Koch

Abstract

Purpose – Generative artificial intelligence (GenAI) constitutes an emerging domain with potential implications for organisational knowledge management (KM). Whilst the absorptive capacity (ACAP) framework is well-established, understanding of how GenAI is applied to the processes within its dimensions remains fragmented. This study mapped and synthesised existing knowledge about GenAI in knowledge management processes, considering the dimensions of absorptive capacity (recognition, acquisition, assimilation, transformation and application).

Design/methodology/approach – A systematic review was conducted using the Scopus and Web of Science databases. The strategy combined terms related to absorptive capacity and knowledge management with specific GenAI descriptors. A total of 126 articles were initially identified, of which 13 were selected for final analysis.

Findings – Four thematic groups emerged: integration with existing systems (31%), as the principal theme; adoption factors (31%), focusing on recognition and application; sharing and application (23%), characterised by high centrality but low density; and knowledge creation (15%), a specialised area. The analysis identified asymmetric development across ACAP dimensions, with concentration in recognition and knowledge application, contrasting with gaps in assimilation.

Originality/value – The study offers a preliminary theoretical contribution by identifying the pattern of dimensional asymmetry in emerging literature on GenAI-mediated ACAP, wherein technological democratisation may challenge assumptions of traditional KM theories. For practice, evidence suggests that organisations may consider ACAP readiness assessment in implementation, prioritising the development of knowledge assimilation capabilities through structured validation protocols.

Keywords Knowledge sharing, Generative artificial intelligence, Knowledge absorptive capacity, Digital transformation, Knowledge management

Paper type Research paper

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1. Introduction

Understanding of the interactions within organisational knowledge management (KM) processes faces challenges related to the codification of tacit knowledge, the speed of information processing and the complexity of knowledge absorption (Pauleen and Wang, 2017; Dávila *et al.*, 2019). Whilst traditional Information Technology systems offer limited support for these processes, recent developments in Artificial Intelligence, particularly the emergence of generative technologies (GenAI), constitutes a potential shift in the technological landscape available to organisations (Kaczorowska-Spychalska *et al.*, 2024). In contrast to conventional artificial intelligence approaches based on rules or simple predictive analyses, GenAI is characterised by its capacity to generate original and contextually relevant content from patterns learnt in extensive data sets (Ramaul *et al.*, 2024a, 2024b), demonstrating attributes of autonomy, explicability and adaptability that position it as a facilitator of KM processes (Darban, 2024).

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Preliminary studies suggest that GenAI can function as an intelligent filtering system in the recognition of strategic knowledge (Leoni *et al.*, 2024), as an automated aggregator in the acquisition of information from multiple sources (Ahn, 2024), as a contextual processor in KM assimilation (Mooradian, 2024), as a recombination mechanism in the creative transformation of existing knowledge (Eisenreich *et al.*, 2024) and as an implementation system in the application of processed knowledge to specific organisational decisions (Rasheed *et al.*, 2024). However, this understanding remains fragmented, indicating the need for theoretical systematisation to facilitate the identification of causal mechanisms, potential benefits and critical challenges of this technological integration.

Understanding the impacts of GenAI on absorptive capacity (ACAP) is particularly complex owing to its multidimensional and interdependent nature. The dimension of value recognition, acknowledged in the literature for the limited human cognitive capacity to process and identify relevant knowledge in vast information repositories (Cohen and Levinthal, 1990), has potential for expansion through GenAI's processing and semantic analysis capabilities (Wang and Zhang, 2024). Simultaneously, the acquisition of external knowledge, historically restricted by cultural and access barriers (Zahra and George, 2002), may be enabled by GenAI's capabilities (Ahn, 2024). However, the assimilation and transformation dimensions present more subtle challenges, as they involve complex cognitive processes of organisational contextualisation and creative recombination (Lichtenthaler, 2016; Yao and Chang, 2017) that may be both enhanced and compromised by GenAI's technological mediation (Sumbal *et al.*, 2024).

The integration of GenAI into KM processes may represent not only an incremental technological innovation, but rather an organisational transformation in the creation, codification and application of knowledge (Zhang *et al.*, 2024). This transformation proposes changes in the division of cognitive labour between human and artificial agents, alterations in power structures based on expertise and redefinitions of organisational decision-making processes (Alavi *et al.*, 2024). Such structural changes indicate the need for systematic investigation that extends beyond isolated technological analyses, integrating perspectives from organisational theory, cognitive psychology and innovation studies (Zhang *et al.*, 2024). Moreover, the emergent and rapidly evolving nature of GenAI technologies indicates the importance of constructing adaptive theoretical frameworks that can accommodate future technological developments, whilst maintaining scientific relevance and practical applicability over time (Lee and Chung, 2024; Mancuso *et al.*, 2025).

This research addresses two gaps in the literature: firstly, it proposes a conceptual synthesis to explain how KM processes are being transformed by GenAI and explains how GenAI technologies impact the dimensions of ACAP. These gaps are supported by the following research questions:

- RQ1.* What are the principal KM processes being transformed by GenAI?
- RQ2.* What mechanisms explain how GenAI affects each dimension of ACAP?
- RQ3.* What are the principal benefits and challenges associated with the integration of GenAI into KM processes?

To address these gaps, a systematic literature review was conducted, following the protocol of Tranfield *et al.* (2003), combining terms related to ACAP and KM with specific GenAI descriptors in the Scopus and Web of Science databases, with the period delimited up to January 2025. The choice of the ACAP framework is justified by its offering of a consolidated structure for understanding ACAP processes (Zahra and George, 2002; Todorova and Durisin, 2007), which enables researchers and organisations to understand how GenAI can contribute to facilitating ACAP in management processes. This article is organised as follows: Section 2 reviews GenAI and theoretical foundations of ACAP; Section 3 details the methodology; Section 4 reports results (bibliometrics, dimensional analysis, strategic mapping); Section 5 discusses theoretical and practical implications and the research agenda; Section 6 concludes with limitations and future directions.

2. Theoretical foundations

GenAI represents an emerging class of systems that generate original content through learning patterns in extensive data sets, differentiating itself from previous generations of AI through its generative capacity rather than merely analytical or predictive approaches (Kaczorowska-Spychalska *et al.*, 2024; Fosso Wamba *et al.*, 2023). Three dimensions distinguish GenAI from previous technological approaches. Firstly, operational autonomy, which enables the execution of complex tasks with reduced human supervision, albeit entailing risks related to the quality of automated decisions (Darban, 2024; Al-Emran *et al.*, 2025). Secondly, explicability represents the aspirational capacity to provide comprehensible justifications for generated outputs, a growing requirement for organisational acceptance and regulatory compliance (Ghobakhloo *et al.*, 2024; Rasheed *et al.*, 2024). Thirdly, contextual adaptability enables adjustment to different organisational demands without extensive reprogramming, whilst requiring integration with existing infrastructures (Wang and Zhang, 2024; Ritala *et al.*, 2024).

Emerging organisational applications encompass three primary domains: advanced natural language processing for contextualised textual interpretation and generation (Raziq *et al.*, 2024; Darban, 2024); creative knowledge generation and recombination that can transcend traditional information retrieval (Khalid *et al.*, 2024; Eisenreich *et al.*, 2024); and decision support through pattern analysis in large volumes of unstructured data (Ardolino *et al.*, 2024; Rasheed *et al.*, 2024).

This study adopts the ACAP framework, originally proposed by Cohen and Levinthal (1990) and refined by Todorova and Durisin (2007), to systematically examine how GenAI may affect KM processes. The model articulates five interdependent dimensions that describe the trajectory through which external knowledge is converted into organisational competitive advantage.

Recognition involves the identification and evaluation of the strategic value of external knowledge through systematic environmental monitoring (Cohen and Levinthal, 1990), enabling organisations to detect opportunities before competitors (Volberda *et al.*, 2010). Acquisition refers to the capture of knowledge identified as valuable via strategic partnerships, collaborations or recruitment of external expertise (Zahra and George, 2002). Assimilation represents a more complex process, involving the analysis, interpretation and internalisation of acquired knowledge, depending on organisational learning routines and integration culture (Todorova and Durisin, 2007; Lichtenthaler, 2016). Transformation combines existing knowledge with new knowledge, generating innovative configurations that augment original sources (Yao and Chang, 2017). Application materialises transformed knowledge into organisational routines, products and processes that generate tangible value (Wang *et al.*, 2015).

The efficacy of these processes depends on specific configurations of contextual, structural and technological factors (Zahra and George, 2002; Todorova and Durisin, 2007). Empirical studies demonstrate that different organisational environments develop distinct ACAP profiles: turbulent environments favour broad-scope capabilities and high flexibility, whilst stable contexts privilege efficiency and narrow focus (Van Den Bosch *et al.*, 1999).

Unlike centralised corporate AI systems that require technical expertise and formal organisational control, democratised GenAI enables individual collaborators to use generative capabilities directly in processes of knowledge recognition, acquisition, assimilation, transformation and application (Sundberg and Holmström, 2023). This direct accessibility may reconfigure assumptions about the *locus* of control in ACAP, potentially challenging established premises in traditional knowledge management theories that assume centralised organisational mediation (Alavi *et al.*, 2024).

3. Procedures and methods

The present study adopts a methodological approach of systematic literature review (SLR), grounded in the guidelines established by Tranfield *et al.* (2003). During the planning

phase, the central objective of the investigation was defined. The data collection strategy focused on the Scopus and Web of Science (WoS) databases, recognised as reference repositories in the scientific community (Paul and Criado, 2020). The review specifically examines GenAI and KM through the ACAP framework, complementing recent studies on AI–KM intersections (Mancuso *et al.*, 2025).

The conduct phase began with the construction of a search strategy directed at the intersection between GenAI and KM. The search string developed combined terms related to ACAP: (“Absorptive Capacity” OR “ACAP” OR “Organisational Learning” OR “Knowledge Management” OR “Knowledge Assimilation” OR “Knowledge Acquisition” OR “Knowledge Transfer” OR “Knowledge Utilisation” OR “Knowledge Transformation” OR “Knowledge Creation” OR “Knowledge Dissemination” OR “Knowledge Application” OR “Knowledge Storage” OR “Knowledge Exploration” OR “Value Recognition”) with specific GenAI descriptors: (“Generative Artificial Intelligence” OR “Generative AI” OR “Generative Model” OR “Generative” OR “Foundation Models” OR “Large Language Models” OR “LLM” OR “Text Generation” OR “Content Generation”).

The systematic search was executed on 7 January 2025, encompassing all documents indexed in the databases up to this date, without restriction of lower temporal limit. In Scopus, the query was applied to the title, abstract and keywords fields (title-abs-key), whilst in Web of Science the Topic field was used, which encompasses title, abstract, author keywords and Keywords Plus. In Scopus, filters were established for the disciplines of Business, Management and Accounting, Economics, Econometrics and Finance, Decision Sciences and Social Sciences. In Web of Science, the categories “Management”, “Business”, “Public Administration” and “Social Sciences Interdisciplinary” were delimited.

Explicit inclusion and exclusion criteria were established to ensure methodological rigour and replicability of the selection process. The inclusion criteria comprised:

- explicit approach to the intersection between GenAI and KM processes;
- focus on organisational contexts;
- full-text availability; and
- publication in the areas of management, business or applied social sciences.

The exclusion criteria included:

- absence of specific mention of GenAI technologies;
- exclusively technical or computational focus, without discussion of organisational implications;
- absence of organisational context, privileging educational, individual or other contexts;
- grey literature, such as white papers and technical reports;
- superficial mentions of GenAI without in-depth theoretical or empirical analysis; and
- misalignment with the scope of knowledge management, addressing only technological or operational aspects without connection to KM processes.

The systematic search identified 126 documents, distributed between 107 records in the Scopus database and 19 in WoS. The results were exported in BibTeX format to the Parsifal systematic review management software, which automatically detected 14 duplicate publications, resulting in 112 articles for screening.

The screening process was conducted collaboratively through the Parsifal platform, in which all authors had shared access for independent evaluation of titles and abstracts. Each article was evaluated by at least two researchers independently, with consensus meetings to resolve any divergences.

The screening resulted in the exclusion of 79 articles based on analysis of titles and abstracts, according to the reasons detailed in [Table 1](#).

Of the 33 articles selected for full-text evaluation, each article was read independently by pairs of researchers, who applied the eligibility criteria and documented the reasons for exclusion. Disagreements were resolved through discussion between the evaluators or, when necessary, with the participation of a third researcher.

The principal reasons for exclusion were: absence of organisational context (11), studies outside the scope of KM (5), superficial analysis of GenAI (2) and insufficient methodological quality (2). This process resulted in a final corpus of 13 articles that fully met the established criteria.

The high exclusion rate (88.4% of unique articles) reflects three characteristics of the field of study. Firstly, the emergent nature of GenAI technologies, with an explosion of publications following the launch of ChatGPT in November 2022, resulting in predominantly exploratory literature, frequently situated in educational contexts (52 excluded articles) or technical-computational contexts, with limited analysis of organisational implications for KM. Secondly, the broad search strategy necessary to ensure exhaustiveness inevitably captured articles that mention GenAI in contexts unrelated to business management. Thirdly, the application of the eligibility criteria prioritised quality and substantive relevance over quantity, excluding studies that, although mentioning GenAI or KM, did not effectively address their intersection in organisational contexts.

3.1 Data analysis and coding

For the qualitative analysis of the 13 selected articles, a structured content analysis process was adopted, grounded in the theoretical model of the five ACAP dimensions proposed by [Todorova and Durisin \(2007\)](#). SciSpace software was used to facilitate the systematic extraction of information from the articles, enabling the identification of relevant textual passages, the marking of key concepts and the generation of structured annotations in each dimension of the ACAP model.

To operationalise the classification of articles according to the five ACAP dimensions, explicit evaluation criteria were established. Each dimension was characterised by specific conceptual elements, textual indicators and identifying keywords. The articles were classified according to the intensity of approach in each dimension (strong, partial or absent), considering the presence, depth and centrality of the defining elements in each study.

The inter-coder agreement index was consistently high, with initial agreement in 89% of dimensional classifications. The residual divergences, concentrated primarily in articles that addressed multiple ACAP dimensions in an integrated manner, were resolved by

Table 1 Screening by title and abstract

<i>Reason for exclusion</i>	N	%
Absence of organisational context	52	65.8
Studies outside the scope of KM	15	19.0
Articles that do not specifically mention GenAI	6	7.6
Studies focused solely on technical aspects of AI	4	5.1
Articles that merely mention GenAI without in-depth analysis	2	2.5
Grey literature	0	0.0
Total	79	100

Source(s): Research data

consensus through discussion grounded in the criteria of [Table 2](#), resulting in unanimous final classification for all 13 articles.

In parallel, a systematic analytical matrix was developed in Microsoft Excel® to organise and synthesise the extracted data. The structure of the matrix included:

- bibliometric elements (title, year, journal, authors, citations);
- methodological aspects (context, area, objectives, approach, data collection and analysis methods, sample characteristics);
- empirical findings;
- conclusions;
- theoretical and practical contributions;
- identified limitations; and
- classification according to the five ACAP dimensions.

The thematic analysis of the coded articles resulted in the identification of four principal thematic groups:

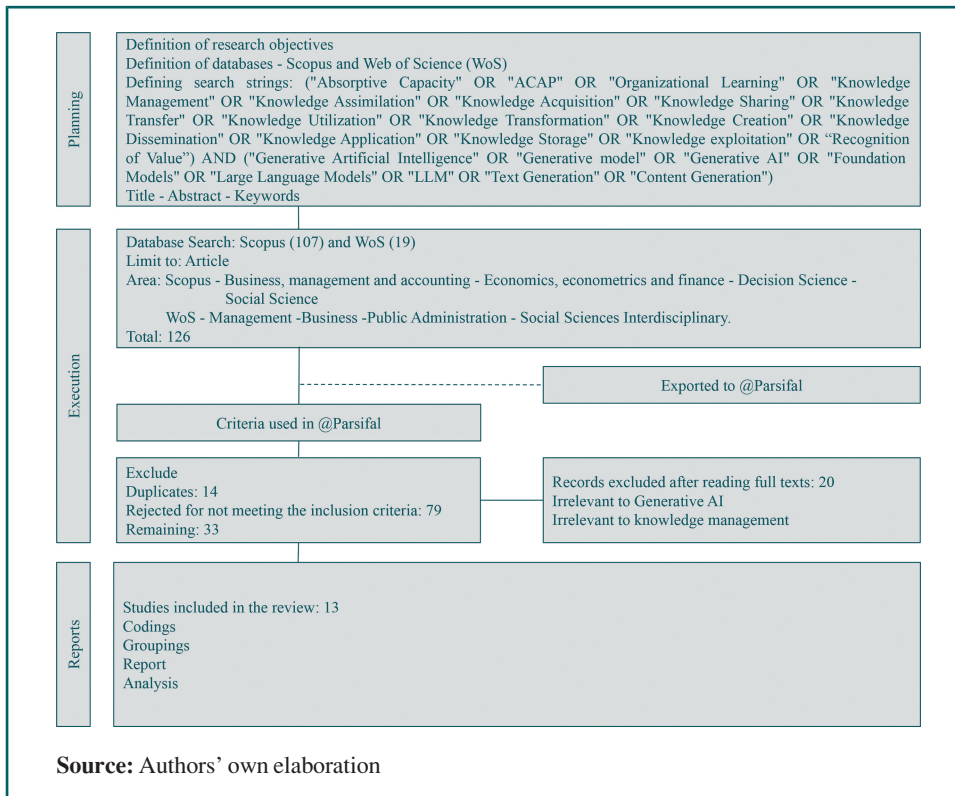
1. GenAI for knowledge creation processes;
2. Integration of GenAI with existing KM systems;
3. Impact of GenAI on knowledge sharing and application; and
4. Factors influencing the adoption of GenAI for KM.

The classification of each article in the ACAP dimensions and thematic groups was validated through consensus between the two coders responsible for each article. The complete detailed research protocol is presented in [Figure 1](#).

Table 2 Coding criteria for absorptive capacity dimensions

<i>ACAP dimension</i>	<i>Classification indicators</i>	<i>Identifying keywords</i>
<i>Recognition</i> Identification and valuation of relevant external knowledge	Identification of external knowledge sources Search mechanisms and environmental scanning Evaluation of strategic value of information Capacity to perceive knowledge opportunities	"identification", "recognition", "search", "scanning", "value recognition", "environmental scanning", "opportunity recognition", "knowledge sources"
<i>Acquisition</i> Obtaining and capturing external knowledge	Processes of capturing external knowledge Channels and acquisition mechanisms Initial internalisation of information Knowledge collection strategies	"acquisition", "capture", "obtain", "gather", "internalization", "knowledge acquisition", "external knowledge", "sourcing"
<i>Assimilation</i> Comprehension and interpretation of acquired knowledge	Analysis and interpretation processes Comprehension and digestion of knowledge Initial cognitive integration Translation and decoding of information	"assimilation", "understanding", "interpretation", "comprehension", "sense-making", "analysis", "digestion", "cognitive integration"
<i>Transformation</i> Conversion and reconfiguration of knowledge	Conversion and combination processes Reconfiguration of existing knowledge Creation of new cognitive schemas Synthesis and integration of diverse knowledge	"transformation", "conversion", "reconfiguration", "combination", "synthesis", "integration", "knowledge creation", "innovation"
<i>Application</i> Utilisation and implementation of knowledge	Practical application of knowledge Implementation and execution Utilisation in organisational processes Generation of results and value Routinisation and institutionalisation	"application", "exploitation", "implementation", "utilization", "use", "deployment", "execution", "operationalization", "routinization"

Source(s): Developed by the authors based on [Todorova and Durisin \(2007\)](#)

Figure 1 Research protocol

4. Results of the research

4.1 Bibliometric profile of scientific production

The bibliometric analysis of the 13 selected studies is presented in [Table 3](#), summarising methodological approaches, main conclusions, theoretical contributions and their relationship to the central theme of the review. This organisation allows for the identification of convergent patterns, theoretical alignments and empirical evidence.

The methodological analysis demonstrates empirical rigour with quantitative emphasis, with PLS-SEM as the dominant approach, complemented by qualitative studies using thematic analysis. From a theoretical perspective, KM theory predominates, whilst contributions emerge from the application of affordance theory, which reconceptualises GenAI's enabling capabilities and from philosophical perspectives on tacit knowledge. Thematically, the studies converge in demonstrating how generative technologies can enhance traditional knowledge processes, address transformations in knowledge-intensive work through creative and conversational mechanisms and evidence GenAI's versatility across diverse contexts.

Convergent patterns emerge from the corpus with implications for theory and practice. Organisational efficiency constitutes a benefit of GenAI adoption, whilst knowledge sharing reveals itself as a mediator of technological transformations, operating in complementarity with GenAI rather than being replaced by it. Dialectical tensions manifest in the opposition between novelty and feasibility in innovative applications and between extractive and enabling approaches in implementation strategies. Concomitantly, decision-making enhancement permeates diverse organisational contexts through simultaneous gains in decisional quality and speed. These convergences suggest strategic imperatives for organisational implementation: primacy of enabling approaches, attention to organisational

Table 3 Detailed bibliometric analysis

Code	Study identification	Methodology and key findings	Theoretical contributions	Relation to review theme
A1	AI-empowered KM processes for decision-making: empirical evidence from worldwide organisations Authors: Leoni et al. (2024) Journal: <i>Journal of Knowledge Management</i> Keywords: Artificial intelligence; decision-making; framework; knowledge management; thematic analysis Navigating virtual teams in generative AI-led learning: the moderation of team perceived virtuality Authors: Darban (2024) Journal: <i>Education and Information Technologies</i> Keywords: AI autonomy; AI explainability; artificial intelligence; human-agent teams; project-based learning; team perceived virtuality	Methodology: qualitative study PLS-SEM Key findings: developed framework interconnecting AI, KM and decision-making through linear and retroactive relationships; Identified 20 factors affecting AI adoption in KM processes; Demonstrated increased efficiency as primary benefit Methodology: multi-level longitudinal study with hierarchical linear modelling (HLM) Key findings: AI explainability significantly enhances knowledge update perceptions; AI autonomy effects contingent on team perceived virtuality; both factors promote knowledge acquisition in virtual learning teams	Integrated framework showing relationships between AI, KM and decision-making; first study simultaneously addressing these three concepts; six-step systematic procedure for managers based on rational decision-making model	Core article demonstrating how generative AI enhances KM processes (knowledge creation, storage, sharing, application) for improved organisational decision-making; Provides empirical evidence from 52 organisations worldwide
A2	Green entrepreneurship success in the age of generative artificial intelligence: the interplay of technology adoption, knowledge management and government support Authors: Wang and Zhang (2024) Journal: <i>Technology in Society</i> Keywords: Generative artificial intelligence; government support; green entrepreneurship; green innovation; knowledge management; resource orchestration theory	Methodology: PLS-SEM Key findings: GAI significantly enhances green knowledge management capabilities; green knowledge management drives green innovation; government support amplifies positive effects; used PLS-SEM and fsQCA for validation	Multi-level longitudinal model integrating human-AI teams and anthropomorphism theories; demonstrates how AI characteristics interact with team virtuality in educational contexts	Explores generative AI (ChatGPT) role in knowledge acquisition and learning processes within teams; shows how AI autonomy and explainability moderate knowledge updating in virtual learning environments
A3	Artificial intelligence and policy making: can small municipalities enable digital transformation? Authors: Koliousis et al. (2024) Journal: <i>International Journal of Production Economics</i> Keywords: Digital transformation; generative AI; policy; SEM; strategic decision making	Methodology: survey/questionnaire PLS-SEM case study Key findings: AI-based strategic decision-making outperforms human decision-making in applicability, efficiency and accuracy; developed three-phase AI framework for policymaking; validated through transportation industry case study	Integrates resource orchestration and knowledge management theories; demonstrates GAI's role in sustainable entrepreneurship; provides dual-method validation (PLS-SEM and fsQCA)	Demonstrates how generative AI adoption enhances knowledge acquisition and application (green knowledge management), leading to innovation and entrepreneurship success in sustainability context
A4			Novel methodological AI-based approach for policymaking through organisational learning theory lens; demonstrates superiority of AI in strategic-level decisions	Shows digital transformation through generative AI platform for strategic policymaking; demonstrates AI's capability to improve organisational learning and knowledge-based decision processes in municipalities

(continued)

Table 3

Code	Study identification	Methodology and key findings	Theoretical contributions	Relation to review theme
A5	Wind of change: how ChatGPT and big data can reshape the knowledge management paradigm? Authors: Sumbal et al. (2024) Journal: <i>Industrial Management and Data Systems</i> Keywords: Big data; ChatGPT; decision making; explicit knowledge; knowledge management; tacit knowledge Generative AI as source of change of knowledge management paradigm Authors: Kaczorowska-Spychalska et al. (2024) Journal: <i>Human Technology</i> Keywords: ChatGPT; digital transformation; generative AI; information and communication technologies; knowledge management	Methodology: case study Key findings: ChatGPT enhances information retrieval, personalised learning, collaborative knowledge sharing and real-time decision support; integration with organisational knowledge base improves decision-making Methodology: Qualitative study survey/questionnaire Key findings: Survey with industry experts reveals generative AI as source of KM paradigm change; identifies benefits, challenges and future trends; digital transformation accelerated by GenAI adoption	Conceptual framework for integrating organisational KM systems with ChatGPT; identifies risks and opportunities of generative AI in KM; proposes mechanism for big data integration Provides industry perspective on generative AI's transformative potential; empirical evidence from experts on KM evolution; framework of GenAI impact on management	Directly addresses how ChatGPT and big data can reshape KM paradigm; explores integration of generative AI with internal organisational knowledge bases for enhanced decision-making Examines how generative AI (ChatGPT) serves as catalyst for KM paradigm shift in digital transformation era; explores current understanding and predicted influence on organisational KM
A7	Revolution or inflated expectations? Exploring the impact of generative AI on ideation in a practical sustainability context Authors: Eisenreich et al. (2024) Journal: <i>Technovation</i> Keywords: Feasibility; generative AI; ideation; novelty; sustainability; sustainable packaging Human resource analytics in the era of artificial intelligence: leveraging knowledge towards organisational success in Pakistan Authors: Rasheed et al. (2024) Journal: <i>Journal of Chinese Human Resources Management</i> Keywords: Artificial intelligence; HR analytics; knowledge sharing; organisational performance; workplace coordination	Methodology: Qualitative study Key Findings: GenAI generates highly novel ideas, occasionally outperforming expert sessions; Trade-off between novelty and feasibility; Context-specific integration needed for optimal results Methodology: Survey/questionnaire PLS-SEM Key findings: Implicit and explicit coordination practices increase knowledge sharing and administrative performance; knowledge sharing mediates relationship; generative AI moderates effects	Comparative effectiveness study of AI vs. expert ideation; practical guidelines for AI integration in innovation processes; identifies barriers at personal and organisational levels Demonstrates mediating role of knowledge sharing in HR analytics; shows moderating effect of generative AI adoption on knowledge coordination and performance	Demonstrates GenAI's role in knowledge creation and innovation processes (ideation phase); shows how AI-generated knowledge compares to expert-generated knowledge in practical sustainability context Explores how generative AI models enhance knowledge sharing systems and administrative performance; links workplace coordination with knowledge management through AI infusion
A8				

(continued)

Table 3

<i>Code</i>	<i>Study identification</i>	<i>Methodology and key findings</i>	<i>Theoretical contributions</i>	<i>Relation to review theme</i>
A9	Creational and conversational AI affordances: How the new breed of chatbots is revolutionising knowledge industries Authors: Ramaul et al. (2024a, 2024b) Journal: <i>Business Horizons</i> Keywords: Chatbots; ChatGPT; conversational AI; generative AI; technology affordances	Methodology: Qualitative study Key Findings: Identified 3 creational affordances (content creation, knowledge acquisition, task automation) and 3 conversational affordances (contextual sensitivity, interactive accessibility, human-AI synergy)	Affordance theory framework for GenAI; introduces meta-affordances (accessibility, accumulation, adaptability); shows how creational and conversational affordances reinforce each other	Comprehensive analysis of how new breed of chatbots (ChatGPT) revolutionises knowledge industries through creational and conversational capabilities; explores knowledge work transformation
A10	Is knowledge management (finally) extractive? – Fuller's argument revisited in the age of AI Authors: Mooradian (2024) Journal: <i>Interdisciplinary Journal of Information, Knowledge and Management</i> Keywords: abductive reasoning; artificial intelligence; automation; knowledge enablement; knowledge management; knowledge work; tacit knowledge	Methodology: PLS-SEM Key findings: Fuller's extractive KM argument remains relevant in AI era; more robust conception of tacit knowledge supports human-enabling view; distinguishes replacement vs. enabling automation strategies	Conceptual analysis of tacit knowledge in KM; philosophical examination of automation strategies; updates fuller's critique for AI age; proposes non-extractive, human-enabling KM approach	Critical analysis of whether KM becomes extractive with AI; examines tacit vs. explicit knowledge in context of generative AI; addresses fundamental question of AI's role in knowledge work
A11	Role of perceived threats and knowledge management in shaping generative AI use in education and its impact on social sustainability Authors: Al-Emran et al. (2025) Journal: <i>International Journal of Management Education</i> Keywords: Generative AI; knowledge Management; social sustainability; T-EESST; TTAT	Methodology: survey/questionnaire PLS-SEM Key findings: Perceived threats negatively impact GenAI use; knowledge acquisition and application drive effective use; GenAI use positively impacts social sustainability	Integrates TTAT and T-EESST theories with KM factors; demonstrates role of knowledge management in GenAI adoption; links AI adoption to social sustainability outcomes	Examines how knowledge management factors (acquisition and application) shape generative AI use in education; shows connection between AI-enhanced KM and social sustainability
A12	Are both generative AI and ChatGPT game changers for 21st-Century operations and supply chain excellence? Authors: Fosso Wamba et al. (2023) Journal: <i>International Journal of Production Economics</i> Keywords: ChatGPT; disruptive technology; emerging technologies; generative artificial intelligence; operations and supply chain management; organisational learning	Methodology: Online survey with descriptive statistics Key Findings: Increased efficiency as significant benefit for adopters and non-adopters; Security and ethical concerns prominent; Organisational learning speeds up GenAI results; Enhances supply chain performance	Organisational learning theory application to GenAI adoption; empirical evidence from UK and USA practitioners; comparative analysis of adopters vs. non-adopters	Investigates GenAI/ChatGPT's transformative impact on operations and supply chain management; shows how organisational learning facilitates knowledge-intensive business transformation through AI

(continued)

Table 3

<i>Code</i>	<i>Study identification</i>	<i>Methodology and key findings</i>	<i>Theoretical contributions</i>	<i>Relation to review theme</i>
A13	AI-powered E-learning for lifelong learners: impact on performance and knowledge application Authors: Ahn H. Y. Journal: Sustainability (Switzerland) (2024) Keywords: academic performance; AI learning; individual impact; job performance; knowledge application; technicality	Methodology: PLS-SEM Key findings: Ease of use, knowledge application and AI learning confidence impact performance and usage frequency; AI tools enhance academic and job performance when properly integrated	PLS-SEM model showing determinants of AI adoption in lifelong learning; links technology usability with knowledge application and learning confidence	Demonstrates how AI-powered e-learning enhances knowledge application and performance; shows importance of user-friendly AI for effective knowledge transfer and learning outcomes
Source(s): Research data				

learning processes and socio-technical knowledge-sharing systems, user-centred design principles privileging explainability and usability and development of capabilities to balance generative potential with operational feasibility constraints.

The temporal distribution of publications reveals the emergent nature of this research domain. As shown in Figure 2, 84.6% ($n=11$) of the articles were published in 2024, with one article in 2023 and one publication from early 2025. This marked concentration in a single year indicates that academic investigation is in its initial stage, with research effectively commencing following the launch of ChatGPT in November 2022, which catalysed widespread organisational experimentation with generative technologies.

This temporal concentration has implications for the interpretation of the results. The temporal depth restricts longitudinal understanding of how GenAI–KM integration evolves as organisations mature in adoption. The observed patterns may reflect initial experimentation rather than stabilised practices. Beyond the rapid acceleration of publications, suggesting that the field is responding to practitioners' demand for guidance, potentially privileging application-oriented research over theoretical development. Finally, the concentration in 2024 implies that the majority of studies were submitted/accepted in late 2023 or early 2024, which means that the empirical data probably capture very early-stage implementations, limiting generalisation to mature deployments.

To examine the relationships between concepts, a co-occurrence network was constructed analysing which keywords appear together in the same articles. Figure 3 presents the keyword co-occurrence matrix for the most frequent terms.

The co-occurrence matrix visualises the relational structure between the 10 principal keywords. The analysis identifies six co-occurrence pairs with maximum strength ($n=2$, each representing 15.4% of the corpus):

1. Knowledge management ↔ Generative AI;
2. Knowledge management ↔ Artificial intelligence;
3. Knowledge management ↔ ChatGPT;
4. Knowledge management ↔ Tacit knowledge;
5. Generative AI ↔ ChatGPT; and
6. Generative AI ↔ Digital transformation.

This pattern presents KM as an absolute hub, connecting with maximum strength to four different keywords that encompass both technological terms (GenAI, AI, ChatGPT) and

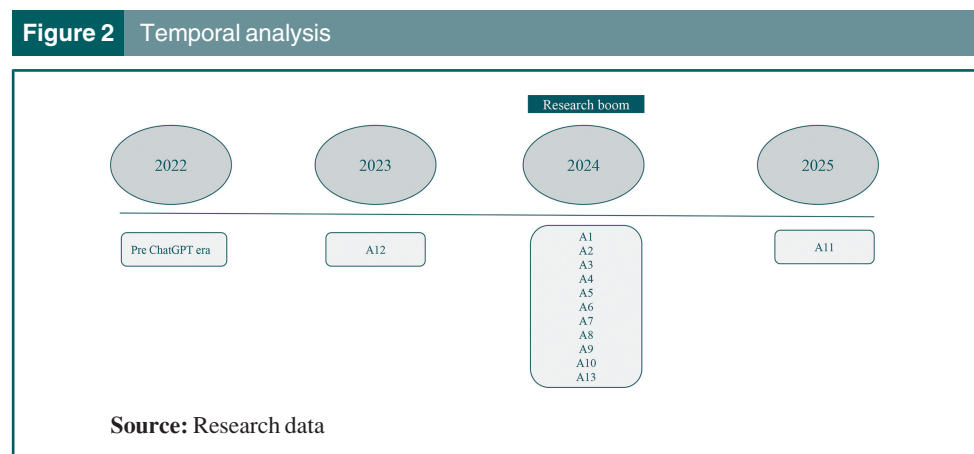
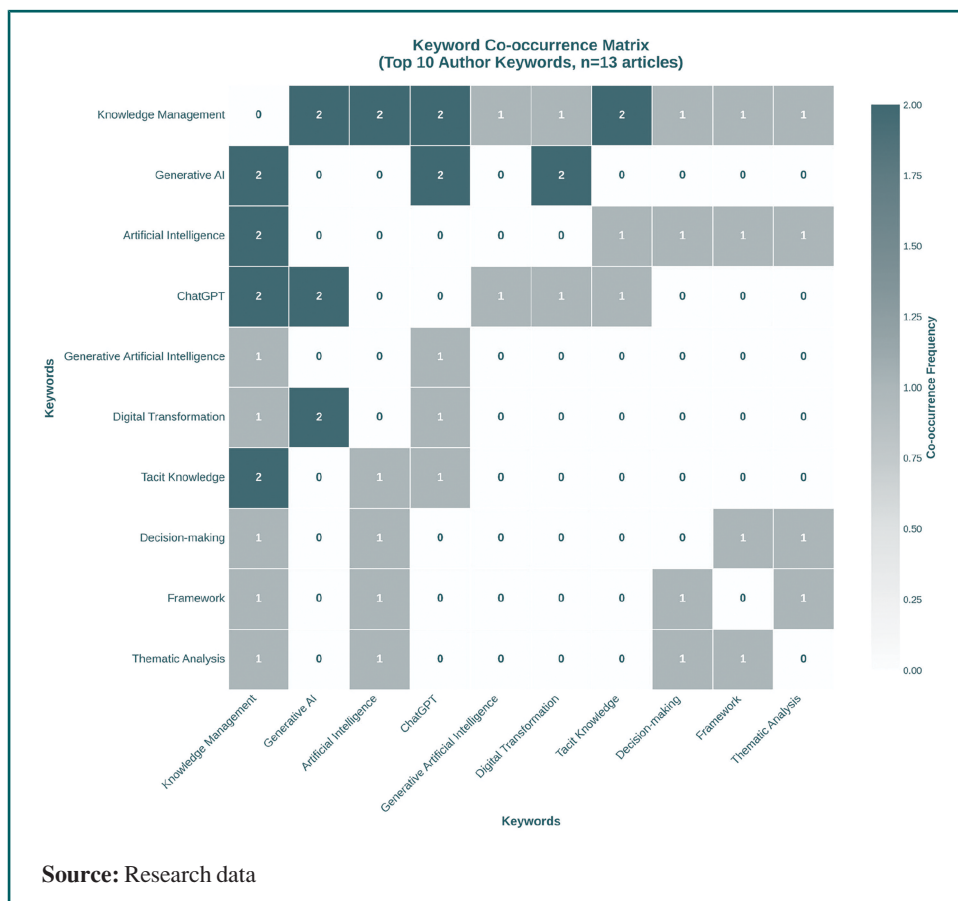


Figure 3 Keyword co-occurrence matrix



process-oriented concepts (tacit knowledge). Moreover, the technological keywords demonstrate strong interconnections amongst themselves (GenAI–ChatGPT, GenAI–digital transformation), evidencing the formation of a technology-focused cluster. In contrast, the peripheral keywords (decision making, framework, thematic analysis) show limited connections ($n \leq 1$), indicating thematic fragmentation.

4.2 Conceptual mapping and thematic analysis

A distinctive aspect of this literature emerges in the emphasis on empirical applications of GenAI in KM processes, contrasting with purely conceptual approaches in emerging fields. Of the 13 articles analysed, 11 (84.6%) report actual organisational implementations or empirical case studies, enabling analysis revealing that researchers examine GenAI not as a theoretical possibility but as a technology actively used in business contexts.

The systematic mapping of articles in relation to the five ACAP dimensions proposed by [Todorova and Durisin \(2007\)](#) presents patterns of coverage and potential theoretical gaps, as demonstrated in [Table 4](#).

The Recognition dimension emerges as one of the most contemplated in the analysed literature, with 12 articles (92.3%) presenting strong or partial coverage, whilst nine studies (69.2%) demonstrate robust treatment. This predominance may be associated with the natural potentialities of GenAI for pattern identification, relevance evaluation and determination of strategic value in vast volumes of information. The reviewed studies suggest that GenAI tools

Table 4 Mapping matrix						
Article	Recognition	Acquisition	Assimilation	Transformation	Application	Specific Thematic
A1	□ Strong	□ Strong	● Partial	● Partial	□ Strong	AI-KM integration for decision-making
A2	□ Strong	□ Strong	● Partial	● Partial	□ Strong	Virtual team learning
A3	□ Strong	● Partial	✗ No	● Partial	□ Strong	Green entrepreneurship and GenAI
A4	● Partial	✗ No	✗ No	● Partial	□ Strong	HR analysis and sharing
A5	□ Strong	● Partial	□ Strong	□ Strong	□ Strong	Paradigmatic transformation via ChatGPT
A6	□ Strong	● Partial	● Partial	□ Strong	● Partial	KM paradigm changes
A7	□ Strong	✗ No	✗ No	□ Strong	□ Strong	AI-assisted ideation processes
A8	□ Strong	● Partial	● Partial	✗ No	□ Strong	Organisational performance and dissemination
A9	□ Strong	● Partial	● Partial	□ Strong	□ Strong	GenAI creation affordances
A10	□ Strong	✗ No	□ Strong	□ Strong	□ Strong	Integration with legacy systems
A11	□ Strong	□ Strong	● Partial	✗ No	□ Strong	Sustainability and implementation
A12	□ Strong	● Partial	□ Strong	□ Strong	□ Strong	Business operations and supply chain
A13	□ Strong	□ Strong	● Partial	✗ No	□ Strong	Team sharing facilitation
Source(s): Research data						

facilitate the identification of strategic knowledge, the recognition of innovation opportunities and the evaluation of the relevance of external information. The predominance of the Recognition dimension finds alignment with the original conceptualisation of [Cohen and Levinthal \(1990\)](#), who positioned this dimension as dependent on the diversity of organisational context, the intensity of R&D activity and the sophistication of the competitive intelligence system. Subsequent empirical studies, including quantitative analyses with Brazilian companies ([Volberda et al., 2010](#)), identify that organisations with greater recognition capacity tend to recognise knowledge opportunities before competitors. [Wang et al. \(2015\)](#) evidenced that recognition capacity acts as a mediator between information systems use and innovative performance, offering an explanation for the natural focus of GenAI literature on this dimension, which presents more evident possibilities for automated filtering and identification.

The Acquisition dimension presents moderate development, with 30.8% of articles demonstrating robust coverage, whilst another six articles offer partial treatment. This distribution suggests that research is still exploring how GenAI facilitates the processes of obtaining and collecting external knowledge previously identified as valuable. The emerging literature in this dimension addresses aspects such as the development of multiple channels of knowledge access, the optimisation of search processes and the establishment of automated information capture mechanisms. However, the moderate

coverage indicates the need for more in-depth investigation into how the technology alters traditional organisational knowledge acquisition processes, particularly in contexts of strategic partnerships and inter-organisational collaborations.

Assimilation constitutes the least explored dimension in the analysed corpus, with only 23.1% of articles demonstrating robust treatment, whilst seven articles offer partial coverage and three do not address this dimension. The scarcity of research in this dimension suggests that the current literature lacks understanding of how GenAI may facilitate (or hinder) the processes of interpretation, analysis and organisational contextualisation of acquired knowledge. This limitation assumes a critical character when considering that effective assimilation requires the development of organisational routines, shared cognitive structures and learning processes that facilitate the internalisation of external knowledge.

The Transformation dimension occupies an intermediate position in the distribution, with 46.2% of articles presenting strong coverage, indicating moderate recognition amongst researchers in this area of GenAI's possible knowledge recombination capabilities. This dimension emerges as potentially significant in innovation processes and the development of new organisational competencies, where the literature suggests the technology's potential to facilitate synthesis activities and create new knowledge configurations that may surpass original sources. Studies in this dimension address aspects such as creative recombination, development of innovative solutions and facilitation of organisational experimentation processes. The moderate coverage suggests that, whilst researchers recognise the transformative capabilities of GenAI, there remains a need for greater understanding of the specific mechanisms by which the technology may facilitate knowledge recombination and synthesis processes in complex organisational contexts.

Application constitutes the most contemplated dimension in the 13 analysed articles, with 84.6% demonstrating strong coverage, evidencing a predominant focus on final results and practical implementations of GenAI. This over-representation indicates that current research is predominantly results-oriented towards tangible outcomes. The reviewed literature in this dimension demonstrates a solid understanding of how GenAI may translate into concrete organisational improvements, from process optimisation to the development of new competitive capabilities. This concentration suggests a pragmatic field approach, in which researchers prioritise the demonstration of the technology's practical value, but also reveals a possible neglect of the intermediate processes that enable these effective applications.

The dimensional analysis of the studies presents an emerging research field characterised by asymmetric developments that suggest both opportunities and challenges for theoretical and practical advancement. The concentration in the recognition and application dimensions, combined with the gap in assimilation, suggests that the current literature better comprehends the extremes of the ACAP process than the intermediate mechanisms. This distribution may reflect both the more evident capabilities of GenAI and possible current conceptual limitations regarding how these technologies actually operate in complex organisational contexts. The future development of the field will potentially depend on the capacity to develop a more balanced and integrated understanding of the five dimensions, particularly through theoretical deepening in the assimilation dimension, which constitutes the connection between external knowledge acquisition and its transformation into distinctive organisational capabilities.

The content analysis identified four principal thematic groups, presented in [Table 5](#), which emerge from the literature, maintaining the previously identified structure, but presenting distinct patterns.

The distribution of the identified thematic groups is aligned with the empirical taxonomies established in the KM literature. Quantitative studies in non-GenAI contexts have identified similar patterns, distinguishing people-centred versus process/infrastructure approaches ([Foss and Michailova, 2009](#)). This convergence between the thematic analysis and

Table 5 Distribution summary

<i>Thematic group</i>	<i>Number of articles</i>	<i>Article codes</i>	<i>% of total</i>
Group 1: GenAI for knowledge creation processes	2	A7, A9	15
Group 2: Integration of GenAI with existing KM systems	4	A1, A5, A6, A10	31
Group 3: Impact of GenAI on knowledge sharing and application	3	A2, A8, A13	23
Group 4: Factors influencing GenAI adoption for KM	4	A3, A4, A11, A12	31
Total	13	A1–A13	100

Source(s): Research data

established empirical evidence suggests conceptual continuity that emerges at the intersection between GenAI and KM, indicating that the identified thematic groups reflect dimensions of KM processes.

Group 1, denominated “GenAI for knowledge creation processes”, represents the smallest proportion of the current literature, with only 15% of articles. This thematic area concentrates on the exploration of ideation and creation processes assisted by GenAI, demonstrating strong alignment with the recognition and transformation dimensions of ACAP. The literature in this group contributes theorisations on technological possibilities for organisational creativity and idea generation mechanisms through recombination of existing knowledge. Although numerically reduced, this group presents high specialisation and internal development, characterising itself as an emerging research area.

Group 2, “Integration of GenAI with existing KM systems”, constitutes the largest thematic category, with 31% of articles, examining how GenAI integrates with established KM systems. This group presents balanced coverage across multiple ACAP dimensions, including value recognition in legacy systems, possible transformation of organisational paradigms and application in strategic decision-making contexts. The dimensional analysis suggests that this group represents the theoretically most mature area.

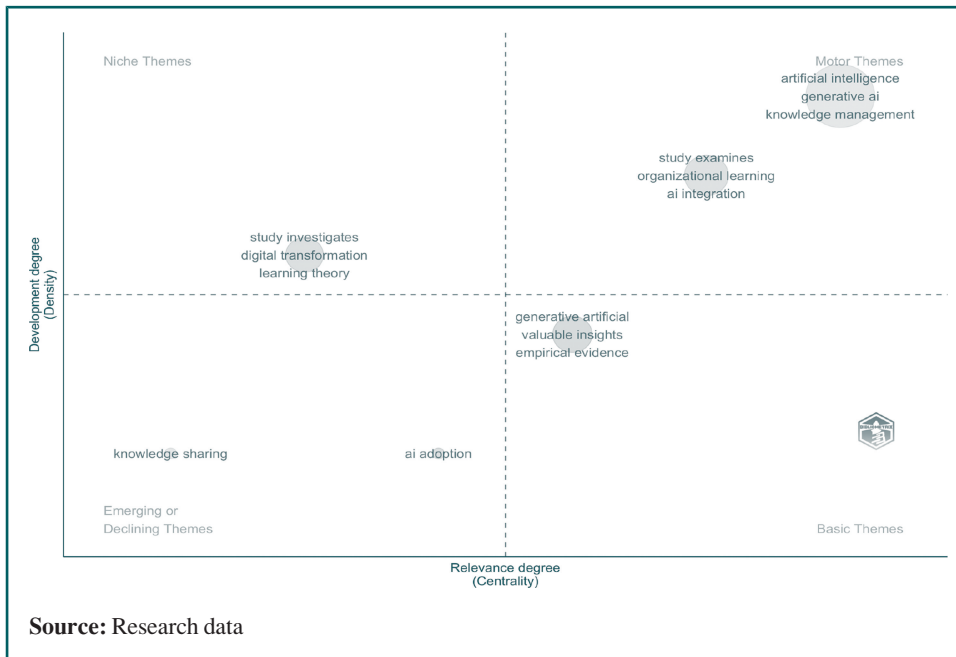
Group 3, “Impact of GenAI on knowledge sharing and application”, encompasses 23% of articles and concentrates on the investigation of possible impacts on organisational knowledge dissemination and utilisation. This group demonstrates focus on the application and recognition dimensions, with particular emphasis on facilitating sharing in virtual teams and its potential impact on organisational performance. However, the dimensional analysis reveals limited theoretical development in the intermediate dimensions of acquisition, assimilation and transformation, indicating opportunities for conceptual development.

Group 4, “Factors influencing the adoption of GenAI for KM”, also represents 31% of articles and investigates the determinants and contexts of GenAI implementation in organisational environments. This group presents an interesting dimensional distribution, with strong presence in recognition, focusing on the identification of adoption opportunities and in application, examining the results of technological implementation. However, similarly to Group 3, it demonstrates limited development in intermediate processes, suggesting that research on adoption is predominantly focused on value identification and final results, with less attention to the processual mechanisms by which adoption effectively occurs.

The strategic analysis of themes based on the identified abstracts reveals the relative position of the different thematic groups identified in terms of centrality (degree of importance) and density (degree of development), as presented in [Figure 4](#).

The motor themes (high centrality and high density) are represented by the concepts “artificial intelligence”, “generative AI” and “knowledge management”, which are positioned in the upper right quadrant of the map. These central themes correspond directly to Group 2 – integration with existing systems, which, with 31% of articles, represents the most mature and influential area of current research. The analysis of the five dimensions reveals that this group presents the most balanced distribution amongst ACAP processes, with strong presence in

Figure 4 Thematic strategic map



recognition (value identification in integration), transformation (paradigmatic changes) and application (practical implementation), functioning as conceptual pillars of the area.

The basic themes (high centrality and low density) include terms such as “generative artificial”, “valuable insights” and “empirical evidence”, positioned in the lower right quadrant. This category corresponds to Group 3 – knowledge sharing and application, representing 23% of articles. The dimensional analysis proposes that this group has a strong focus on recognition and application, but limited development in intermediate dimensions, indicating opportunities for conceptual development, particularly in the acquisition, assimilation and transformation processes.

The specialised themes (low centrality and high density) are represented by “study investigates”, “digital transformation” and “learning theory”, located in the upper left quadrant. These themes correspond to Group 1 – knowledge creation, which, although representing only 15% of articles, demonstrates high specialisation and internal development. The analysis of the five dimensions shows strong alignment with recognition (identification of creative opportunities) and transformation (creative recombination), representing a specialised area with expansion potential, especially if creative processes become more central to organisational applications of GenAI.

Emerging or declining themes (low centrality and low density) include “knowledge sharing” and “AI adoption”, positioned in the lower left quadrant. This category corresponds to Group 4 – adoption factors, which, despite representing 31% of articles, still presents low theoretical development and limited influence. The dimensional analysis reveals an interesting concentration in recognition and application, but limited development in intermediate dimensions, suggesting that, whilst there is interest in adoption factors, research still lacks an in-depth understanding of the processes by which adoption effectively occurs.

Research such as that of [Sumbal et al. \(2024\)](#) addresses the challenges of GenAI implementation in KM, mentioning that organisations face difficulties in understanding how these technological competencies can be integrated into existing KM systems. This finding is corroborated by [Ghobakhloo et al. \(2024\)](#), who discuss the “perceived threats” and barriers in the implementation of AI technologies in organisational contexts, focusing on adoption challenges and KM.

The trajectory analysis reveals distinct evolution patterns, to understand how fields may evolve strategically, as represented in [Figure 5](#).

The consolidation trajectory characterises the development of Group 2 – integration with existing systems, positioned as a motor theme with high centrality and high density. This trajectory emphasises the evolution of motor themes towards greater theoretical sophistication through the development of standardised implementation frameworks, integration with established KM theories and empirical validation through longitudinal studies. The natural movement towards consolidation reflects the theoretical maturity of the group and its central position in the field, suggesting that future investments should focus on conceptual refinement and the development of practical tools that facilitate organisational implementation. In contrast, the emergence trajectory represents the development of Group 1 – knowledge creation, classified as a specialised theme with expansion potential for conventional organisational applications through the development of creativity measurement frameworks and integration with innovation management practices. Whilst currently representing a specialised niche, this group possesses significant growth potential, especially considering that creative processes may become increasingly central to organisational applications of GenAI.

Figure 5 Development trajectories of thematic groups

☐ Consolidation Trajectory MOTOR THEMES	☐ Emergence Trajectory SPECIALISED THEMES
Evolution of motor themes towards greater theoretical sophistication and practical implementation frameworks. ☐ Primary Group: Group 2: System Integration Key Elements: • Development of standardised implementation frameworks • Integration with established KM theories • Empirical validation through longitudinal studies • Industry-specific adaptation models • Policy development for organisational adoption	Movement of specialised themes towards central relevance as creative applications become critical. ☐ Primary Group: Group 1: Knowledge Creation Key Elements: • Expansion into mainstream organisational applications • Development of creativity measurement frameworks • Integration with innovation management practices • Cross-industry adoption studies • Theoretical bridges to entrepreneurship literature
☐ Development Trajectory EMERGING THEMES	☐ Expansion Trajectory BASIC THEMES
Evolution of emerging themes towards greater centrality through systematic adoption frameworks. ☐ Target Group: Group 4: Adoption Factors Key Elements: • Multi-level adoption frameworks (individual/organisational) • Cultural and contextual adaptation models • Change management for AI implementation • Resistance mitigation strategies • Success factors identification and validation	Development of basic themes towards greater theoretical density through enhanced sharing mechanisms. ☐ Target Group: Group 3: Sharing & Application Key Elements: • Advanced behavioural models for knowledge sharing • Collaborative platform development • Performance measurement in virtual teams • Integration with organisational learning theories • Cross-cultural knowledge transfer models

Source: Research data

The expansion trajectory characterises the development of Group 3 – knowledge sharing and application, positioned as a basic theme with high centrality but low density, demonstrating the need for development of advanced behavioural models for knowledge sharing and collaborative platforms. This trajectory indicates that, whilst the group possesses recognised central relevance, it still requires substantial conceptual deepening to realise its full potential, focusing on mechanisms that facilitate effective dissemination of organisational knowledge. Simultaneously, the development trajectory represents the evolution of Group 4 – adoption factors, classified as an emerging theme, which requires multi-level adoption frameworks, cultural adaptation models and change management strategies for its implementation. This trajectory indicates that, whilst it represents an important area of investigation, it still lacks theoretical development and influence in the field, with emphasis on understanding the processes by which organisations implement GenAI technologies.

Theoretical development should focus on the integration of four simultaneous streams encompassing consolidation theories, creative emergence, collaborative expansion and adoption development, whilst practical applications need to include specific implementation guides for each trajectory.

5. Discussion of results

The analysis of organisational challenges and advantages identified in the literature, when examined through the lens of the five ACAP dimensions, reveals patterns suggesting where barriers and opportunities are most pronounced in GenAI implementation, as shown in [Figure 6](#).

The analysis of organisational challenges and advantages identified in the literature, when examined through the ACAP dimensions, presents patterns that elucidate where barriers

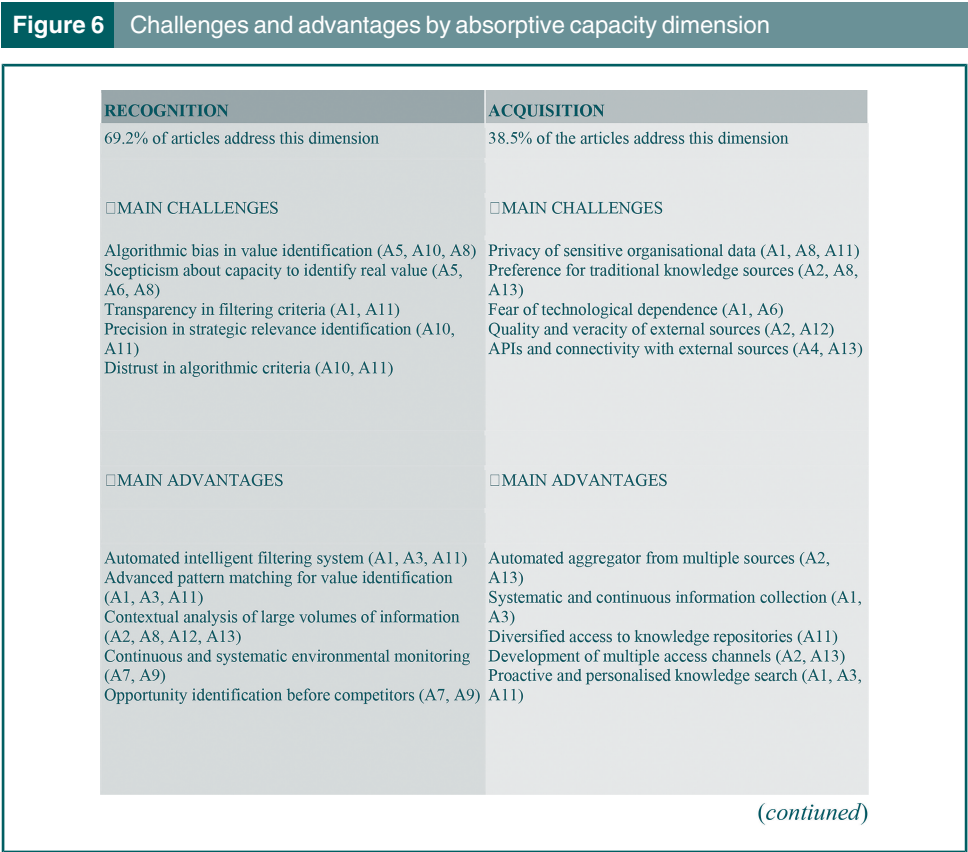


Figure 6

ASSIMILATION	TRANSFORMATION
23.1% of articles address this dimension	53.8% of articles address this dimension
□MAIN CHALLENGES Biased interpretation of organisational contexts (A10, A13) Distrust in automated interpretation (A10, A13) Security in tacit knowledge encoding (A5, A6) Standardisation of knowledge formats (A5, A12) Contextual interpretive precision (A5, A12) □MAIN ADVANTAGES Advanced contextual processor (A5, A10, A12) Automatic interpretation and categorisation (A5, A10, A12) Analysis and processing of complex knowledge (A5, A10) Transformation of information into usable knowledge (A5, A10) Efficient organisational contextualisation (A12)	□MAIN CHALLENGES Intellectual property of creative recombinations (A7, A9) Fear of losing creative control (A7, A9) Transparency in automated creative processes (A6, A10) Originality vs. reproduction of existing patterns (A6, A10) Resistance to work paradigm changes (A6, A10) □MAIN ADVANTAGES Advanced creative recombination engine (A5, A6, A7, A9, A10, A12) Generation of new combinations and applications (A5, A7, A9, A10) Automated creative and analogical processes (A7, A9) Facilitation of diverse knowledge integration (A6, A12) Synthesis and creation of innovative configurations (A1, A5, A10)
APPLICATION	
76.9% of articles address this dimension	
□MAIN CHALLENGES Accountability of automated decisions (A12, A4, A1) Resistance to decision automation (A1, A11, A12) Auditability of AI recommendations (A8, A11) Concern about human role substitution (A4, A8) Compatibility with implementation tools (A4, A8) □MAIN ADVANTAGES Automated implementation system (A1, A5, A8, A13) Translation of knowledge into specific actions (A1, A5) Advanced decision-making support (A1, A5, A8, A13) Improvement of products, processes and services (A3, A11, A4, A12) Integration with existing decision systems (A2, A7, A9, A10)	

Source: Research data

and opportunities are most pronounced in GenAI implementation. In the recognition dimension, which demonstrates 92.3% coverage in the literature, the principal challenges concentrate on issues of algorithmic bias in value identification (Leoni *et al.*, 2024; Kaczorowska-Spychalska *et al.*, 2024) and organisational scepticism regarding technological competence for strategic discernment (Sumbal *et al.*, 2024; Darban, 2024), in addition to transparency problems in filtering criteria that compromise the reliability of the process (Koliousis *et al.*, 2024; Rasheed *et al.*, 2024). However, the advantages are substantial, as evidenced by studies documenting automated intelligent filtering systems capable of processing volumes of information impossible to manage humanly (Leoni *et al.*, 2024; Wang and Zhang, 2024) and the capacity to identify opportunities before competitors (Eisenreich *et al.*, 2024; Ramaul *et al.*, 2024a, 2024b).

In the acquisition dimension, with 30.8% coverage, challenges emerge related to the privacy of sensitive organisational data (Koliousis *et al.*, 2024; Rasheed *et al.*, 2024), preference for traditional knowledge sources (Ahn, 2024; Zhang *et al.*, 2024) and fear of technological dependence (Leoni *et al.*, 2024; Kaczorowska-Spychalska *et al.*, 2024), whilst advantages include automated aggregation from multiple sources (Ahn, 2024; Zhang *et al.*, 2024) and the development of diversified channels of knowledge access (Leoni *et al.*, 2024; Wang and Zhang, 2024). Assimilation, as the least explored dimension with 23.1% coverage, faces more complex challenges. Studies highlight biased interpretation of organisational contexts (Koliousis *et al.*, 2024; Zhang *et al.*, 2024) and technological limitations in understanding specific nuances (Sumbal *et al.*, 2024; Alavi *et al.*, 2024), but also identify potential for contextual processing (Koliousis *et al.*, 2024; Alavi *et al.*, 2024) and transformation of information into usable knowledge (Sumbal *et al.*, 2024; Koliousis *et al.*, 2024).

In transformation, with 46.2% coverage, challenges concentrate on issues of intellectual property related to creative recombinations, fear of losing creative control (Eisenreich *et al.*, 2024; Ramaul *et al.*, 2024a, 2024b) and tensions between originality and reproduction of existing patterns (Kaczorowska-Spychalska *et al.*, 2024; Koliousis *et al.*, 2024), revealing complexities inherent to artificial creativity. Advantages, however, are extensively documented, including advanced creative recombination mechanisms (Sumbal *et al.*, 2024; Kaczorowska-Spychalska *et al.*, 2024; Eisenreich *et al.*, 2024; Ramaul *et al.*, 2024a, 2024b; Koliousis *et al.*, 2024; Alavi *et al.*, 2024), generation of new combinations and applications (Sumbal *et al.*, 2024; Eisenreich *et al.*, 2024; Ramaul *et al.*, 2024a, 2024b; Koliousis *et al.*, 2024) and facilitation of the integration of diverse knowledge (Kaczorowska-Spychalska *et al.*, 2024; Alavi *et al.*, 2024). Application, as the most contemplated dimension with 84.6% coverage, faces challenges related to accountability for automated decisions (Alavi *et al.*, 2024; Rasheed *et al.*, 2024; Leoni *et al.*, 2024), resistance to decision automation (Leoni *et al.*, 2024; Koliousis *et al.*, 2024; Alavi *et al.*, 2024) and compatibility problems with existing tools (Rasheed *et al.*, 2024; Koliousis *et al.*, 2024), reflecting tensions between technological efficiency and organisational control. Simultaneously, the literature documents advantages through automated implementation systems (Leoni *et al.*, 2024; Sumbal *et al.*, 2024; Koliousis *et al.*, 2024; Zhang *et al.*, 2024), decision-making support (Leoni *et al.*, 2024; Sumbal *et al.*, 2024; Koliousis *et al.*, 2024; Zhang *et al.*, 2024) and effective improvement of organisational products, processes and services (Wang and Zhang, 2024; Koliousis *et al.*, 2024; Rasheed *et al.*, 2024; Alavi *et al.*, 2024).

The integrated research indicates that challenges manifest differently in each dimension, requiring specific approaches for each stage of ACAP. Ethical and security issues, mentioned in 76.9% of articles, vary from biases in value identification to accountability for decisions, indicating the need for ethical frameworks that consider the specificities of each dimension. Organisational resistance, present in 53.8% of articles, manifests from scepticism regarding technological capabilities to fear of losing creative control, suggesting the need for differentiated change management strategies.

Quality and reliability challenges, identified in 61.5% of articles, evolve from precision in value identification to implementation effectiveness, indicating that different dimensions require specific metrics and quality controls. This dimensional analysis offers an understanding for the development of implementation strategies that maximise benefits whilst mitigating specific risks.

The identification of challenges by dimension corroborates the conclusions of empirical studies demonstrating how different aspects of ACAP require differentiated management approaches. [Van Den Bosch *et al.* \(1999\)](#) evidenced that, in turbulent knowledge environments, companies tend to develop ACAP with specific characteristics: low efficiency, broad scope and high flexibility, contrasting with stable environments, where they develop capabilities with high efficiency, restricted scope and low flexibility. This finding offers a theoretical explanation for the asymmetric distribution encountered, in which GenAI literature concentrates on dimensions with more evident technological affordances (recognition and application) to the detriment of more complex cognitive processes (assimilation).

5.1 Theoretical implications and research agenda

This study offers three principal theoretical contributions by demonstrating that the ACAP model, whilst robust, requires adaptations when applied to GenAI-mediated contexts.

The first contribution resides in the identification of uneven development between ACAP dimensions when organisations adopt generative technologies. Whilst traditional theory assumes relatively balanced progression between dimensions ([Zahra and George, 2002](#); [Todorova and Durisin, 2007](#)), the findings reveal a leap-frog pattern where organisations rapidly develop capabilities in dimensions with clear technological affordances whilst delaying in cognitive-social processes. This asymmetry suggests that technological mediation alters ACAP development trajectories, favouring automatable dimensions to the detriment of interpretative processes.

The second contribution differentiates GenAI from traditional AI systems in terms of technological democratisation. Whilst previous AI implementations require centralised organisational control and technical expertise, GenAI enables direct use by knowledge workers without intermediation ([Alavi *et al.*, 2024](#)). This democratisation challenges established KM theories that assume organisational control over absorption mechanisms. The identification of two distinct thematic communities (technological versus organisational) evidences tension between individual autonomy and organisational governance.

The third contribution relates to the fragmentation of the emerging field. Bibliometric analysis revealed high terminological dispersion and extreme temporal concentration, indicating an initial stage of theoretical consolidation. This fragmentation, typical of emerging domains ([Tranfield *et al.*, 2003](#)), simultaneously underlines the need for organising frameworks and the opportunity for theoretical development that integrates currently dispersed perspectives.

Based on the identified gaps and asymmetries, five priority areas for future research are proposed to advance theoretical understanding and practical application of GenAI-KM integration. [Table 6](#) synthesises these areas, key questions and methodological approaches.

As presented in [Table 6](#), longitudinal research becomes imperative to understand whether the observed dimensional asymmetries represent a transitory or structural pattern. Comparative analyses across industries, cultures and organisational sizes will enable the identification of boundary conditions for the applicability of the findings. Finally, examination of the “dark side” will balance technological optimism with realistic understanding of risks and long-term negative consequences.

Table 6 Principal research areas and questions for GenAI–KM integration

Research area	Research gap	Key questions	Methodological approaches
Assimilation processes	23.1% coverage versus 92.3% Recognition – gap in understanding validation and integration of opaque AI knowledge	- How do organisations validate GenAI knowledge? - What trust mechanisms emerge? - How do cognitive biases affect assimilation?	- Qualitative case studies - Experimental designs - Process tracing
Longitudinal ACAP development	Cross-sectional focus limits understanding of temporal evolution and maturation patterns	- Do asymmetries persist over time? - What is the optimal development sequence? - How do interventions affect trajectories?	- Longitudinal field studies - Event studies - Time-series analysis
Comparative contextual analysis	Understanding of contingency factors shaping GenAI–KM integration	- Industry-specific patterns? - Size and maturity effects? - Cultural influences on trust/assimilation?	- Multi-industry surveys - Cross-cultural studies - Configurational analysis
Mechanism-Focused investigation	Descriptive studies dominate; causal mechanisms underexplored	- Which GenAI features affect which dimensions? - What micro-processes mediate effects? - Boundary conditions?	- Experiments - Mechanism analysis - Mixed methods
Dark side examination	Positive outcomes bias; risks and negative consequences underexamined	- De-skilling effects? - Bias propagation? - Strategic vulnerabilities?	- Critical case studies - Risk assessments - Longitudinal tracking
Source(s): Prepared by the authors			

5.2 Implications for practice

For managers and organisations seeking to integrate GenAI into knowledge processes, this study offers practical indications at three principal organisational levels:

1. Strategic level. Before implementing GenAI, organisations need to conduct a readiness assessment, identifying existing capabilities and gaps. Given the identification of Assimilation as the least developed dimension in the literature and most complex in practice, this dimension requires particular attention. It is necessary to begin with Recognition tasks where the technology demonstrates maturity; invest in the development of Assimilation capabilities through validation protocols and employee training; establish Transformation processes for creative recombination; and only then scale Application across organisational units. This sequential progression enables gradual construction of capabilities for effective absorption of GenAI-mediated knowledge.
2. Tactical level. Effective implementation requires employee training to work with GenAI. Given the complexity of assimilation, priority should be given to training programmes developing capabilities to: critically evaluate AI-generated content, detect “hallucinations” and inaccuracies, contextualise generic outputs to specific organisational situations and integrate GenAI insights with existing domain expertise. Complementarily, establish structured validation processes with different levels of rigour according to criticality: automated validation for low-risk applications, peer review for medium-risk applications and expert validation for strategic or high-impact decisions. These practices mitigate identified risks related to reliability and accountability whilst enabling capture of technological benefits.
3. Operational level. Define specific use cases by ACAP dimension aligned with organisational maturity. For recognition: competitive intelligence, technological prospecting, customer insight mining. For acquisition: multi-source aggregation, expert knowledge capture, facilitation of external collaborations. For assimilation: expert validation support, contextualisation assistance, knowledge gap analysis. For transformation: brainstorming and idea generation, problem reframing, scenario development. For application: decision support, process documentation, training content creation. Begin with use cases in high-readiness dimensions and expand gradually as the organisation develops capabilities in remaining dimensions.

Complementarily, implement governance that balances individual autonomy in GenAI use with organisational controls necessary for quality, ethics and strategic alignment. This governance addresses issues of privacy, intellectual property, algorithmic bias and accountability for decisions, establishing clear boundaries between permitted experimentation and applications requiring formal approval.

6. Final considerations

This study presented an analysis of research relating GenAI to KM processes. Advancing the literature on the model that articulates the five dimensions, value recognition, acquisition, assimilation, transformation and knowledge exploitation in the ACAP process, it identified that technological democratisation may challenge the assumptions of traditional KM theories.

The findings demonstrate that GenAI emerges both as a technological facilitator and as a potential solution to overcome the structural and cultural barriers identified in previous empirical studies. Evidencing that ACAP readiness in implementation, prioritising the development of knowledge assimilation capabilities through structured validation protocols, can be strategic in organisational practices.

It further identifies that the emergent nature of GenAI results in still limited literature, restricting the depth of available analyses. In this regard, future research may explore the development of prescriptive frameworks that guide managers in the effective implementation of GenAI, considering the specificities of each ACAP dimension, leveraging empirical evidence that different dimensions require differentiated management approaches.

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